

Technology Stack

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3, Bootstrap, JavaScript</i>	<i>Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Flask handles user requests, performs data preprocessing, loads the trained machine learning model, and returns prediction results efficiently.</i>
3	Database / Data Storage	<i>CSV Dataset / IBM Cloud Object Storage</i>	<i>Stores historical applicant data used for training and testing the machine learning models..</i>
4	Cloud / Hosting / Deployment	<i>IBM Watson Machine Learning, IBM Cloud</i>	<i>Deploys the trained machine learning model on the cloud for scalable, real-time credit card approval predictions.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Enables version control, team collaboration, project tracking, and source code management.</i>

6	Third-Party APIs / Other Tools	<i>Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib</i>	<i>Used for data preprocessing, model training, evaluation, visualization, and implementing classification algorithms for credit card approval prediction.</i>
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